Sahaj Bhadja

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Summary

Software engineer completing an MSCS at USC with hands-on **C/C++**, robotics simulation, and integration-testing experience across electromechanical and sensor-driven systems. Skilled in bench validation, modular code reviews, technical documentation, and improving firmware quality, reliability, and traceability in fast-paced engineering teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Operating Systems, Analysis of Algorithms, Computer Networks

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Operating Systems, Data Structures, OOP, Software Engineering

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Contributed to verification-style validation pipelines and reconciliation checks for production systems, documenting software requirements, design decisions, and test outcomes while maintaining **99%+ reliability** across **5K+ monthly jobs**.
- Partnered with cross-functional engineers on system integration troubleshooting, improving modular service behavior and performance through indexed queries that reduced dashboard latency from **1.2s** to **350ms**.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented concurrent, fault-tolerant processing pipelines with structured job-state transitions and isolation, supporting real-time instructor workflows and **10K+ daily requests** without blocking user-facing control interfaces.
- Participated in code reviews and design discussions, writing automated test coverage and developer documentation that reduced manual turnaround by **60%** while learning maintainable modular engineering practices from senior teammates.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built closed-loop robotics simulation pipelines processing **100K+ trajectory logs** at **10Hz**, integrating sensor-context signals and electromechanical motion planning that improved long-horizon stability by **15%** in benchmark rollouts.
- Supported hardware-adjacent system integration and debugging across simulation environments, collaborating with researchers on modular design, test-result documentation, and bench-style validation of control behavior before deployment.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Assisted system-level integration and bench testing through automated VM workflows, nightly Jenkins jobs, and **20+ PyTest** suites validating executables across **Linux**, Windows, and macOS under Agile code-review practices.
- Troubleshot cross-platform integration defects with senior engineers, improving API response times by **30%** and pipeline efficiency by **68%** while documenting test artifacts, failure logs, and traceable release workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Integrated real-time control and monitoring interfaces over WebSocket-driven updates across distributed services, supporting electromechanical-style operational workflows syncing **2M+ records** with **95%+** successful build validation.
- Collaborated on modular service boundaries and Docker-based deployment automation, cutting release cycles by **50%** while supporting system integration, debugging, and performance tuning across billing and operations modules.

Skills

Embedded & Systems: C, C++, embedded C/C++, microcontrollers (familiar), ARM Cortex-M (familiar), bare-metal (familiar), RTOS (familiar), SPI (familiar), I2C (familiar), UART (familiar), CAN (familiar), USB (familiar), motor control (familiar), sensors, robotics, electromechanical systems, hardware bring-up (familiar), real-time systems
Quality & Tools: verification and validation, bench testing, system-level testing, technical documentation, IEC 62304 (familiar), FDA (familiar), code reviews, modular design, traceability, Python, Linux, Git, Jenkins, CI/CD, pytest, debugging, Agile

Projects

ControlsSim | *C++, Motor Control, Sensors, UART-Style Interfaces*
- Built a **C++** motor-control and sensor-feedback simulation harness modeling command/response timing across electromechanical states, supporting bench-level validation of control logic, edge cases, and modular firmware-style modules.

MedDevice Test Systems | *Integration Testing, Bench Validation, Regulated Documentation*
- Contributed to firmware verification and bench testing for medical-device prototypes, documenting requirements, design decisions, and test results with traceability artifacts aligned with regulated quality and handoff practices.

CPU-scheduling Simulator Live | *C++, Real-Time Scheduling, OS Internals*
- Implemented preemptive and non-preemptive scheduling algorithms in **C++** across **1K+** process combinations, applying real-time systems concepts that reduced debugging time by **20%** through performance-focused validation workflows.